ATTESTATION ◆ ATTESTATO ◆ ATESTACIÓN ◆ BESCHEINIGUNG ◆ ATTESTATION

Product Service

Compliance Document

No. D 115067 0077 Rev. 00

Holder of Certificate: **Xiamen Kehua Digital Energy Tech Co., Ltd.**
Room 208-38, Hengye Building
No. 100 Xiangxing Road
Torch High-tech Zone
(Xiangan) Industrial Zone
361115 Xiamen
PEOPLE'S REPUBLIC OF CHINA

Product: **Converter (Energy Storage Inverter)**

Model(s): **BCS1000K-C-HUD, BCS1250K-C-HUD**

Parameters: See page 2

Tested according to: EN 50549-2:2019/A1:2023
EN 50549-10:2022

This Compliance document confirms the compliance with the listed standards on a voluntary basis. It refers only to the sample submitted for testing and certification and does not certify the quality or safety of the serial products. For details see: www.tuvsud.com/ps-cert

Test report no.: 64290253033901

Date, 2025-07-31

Kennen Wang

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Parameters:

Model	BCS1000-C-HUD	BCS1250K-C-HUD
Battery input/output parameters		
Battery type	Lithium-ion / Lead-acid	
Maximum voltage [V_{DC}]	1500	
Battery rated voltage [V_{DC}]	1250	
Battery voltage range [V_{DC}]	1160-1500	
Maximum charge power [kW]	1078.0	1347.5
Maximum discharge power [kW]	1122.4	1403.0
Maximum charge current [A_{DC}]	1078.0	1347.5
Maximum discharge current [A_{DC}]	1122.4	1403.0
Grid terminal input/output parameters		
Rated output voltage [V_{AC}]	3P+PE, 690	
Rated output frequency [Hz]	50	
Rated output current [A_{AC}]	836.8	1046.0
Maximum continuous output current [A_{AC}]	920.5	1150.6
Rated output active power [kW]	1000	1250
Maximum output active power [kW]	1100	1375
Maximum output apparent power [kVA]	1100	1375
Power factor range	0.9 inductive(under-excited) to -0.9 capacitive(over-excited)	

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Evaluated protection function and operational capabilities

Clause(s) / subclause(s) of EN 50549-2:2019+ A1:2023	Applicable clause(s) / subclause (s) of EN 50549-10:2022	Remarks, optional modes and constraints	Verdict
4.4.2 Operating frequency range	5.2.1 Frequency operating range	--	Pass
4.4.3 Minimal requirement for active power delivery at underfrequency	5.2.1 Frequency operating range	--	Pass
4.4.4 Continuous operating voltage range	5.2.2 Voltage operating range	--	Pass
4.5.2 Rate of change of frequency (ROCOF) immunity	5.3.1 Immunity to disturbances - Rated of change of frequency (ROCOF)	--	Pass
4.5.3.2 Generating plant with nonsynchronous generating technology	5.3.3 Immunity to disturbances - Fault ride through, over-voltage (OVRT) and under-voltage (UVRT)	--	Pass
4.5.4 Over-voltage ride through (OVRT)	5.3.3 Immunity to disturbances - Fault ride through, over-voltage (OVRT) and under-voltage (UVRT)	--	Pass
4.5.5 Phase jump immunity	5.3.2 Phase jump	--	Pass
4.6.1 Power response to overfrequency	5.4 Active response to frequency deviation	--	Pass
4.6.2 Power response to underfrequency	5.4 Active response to frequency deviation	--	Pass
4.7.2.2 Voltage support by reactive power, Capabilities	5.5.1 Power capabilities assessment	--	Pass
4.7.2.3 Voltage support by reactive power, Control modes	5.5.2 Voltage support by reactive power - test to determine the reactive power control modes	Q setp Q(U) Cos φ setp Cos φ (P) Q (P)	Pass
4.7.2.3.2 Set point control modes	5.5.2.3 Verification procedure for set point control	Q setp	Pass
4.7.2.3.3 Voltage related control modes	5.5.2.4 Verification procedure for voltage related control mode for reactive power Q(U)	Q(U)	Pass
4.7.2.3.4 Power related control mode	5.5.2.5 Verification procedure for power related control modes for reactive power	Cos φ (P) and Q (P)	Pass
4.7.3 Voltage related active power reduction	5.6 Voltage related active power reduction - P(U)	P(U)	Pass

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4.7.4.2.1 Voltage support during faults and voltage steps - General	5.3.3 Immunity to disturbances - Fault ride through, over-voltage (OVRT) and under-voltage (UVRT)	--	Pass
4.7.4.2.1.2 Voltage support during faults and voltage steps - Optional Modes	5.3.3 Immunity to disturbances - Fault ride through, over-voltage (OVRT) and under-voltage (UVRT)	--	N/A
4.7.4.2.2 Zero current mode for converter connected generating technology	5.3.3 Immunity to disturbances - Fault ride through, over-voltage (OVRT) and under-voltage (UVRT)	--	Pass
4.9.3 Requirements on voltage and frequency protection	5.8.5 Verification procedure for generating plants to be connected to a MV distribution network	External interface protection for generating plants to be connected to the MV network according to EN 50549-2:2019+A1:2023 must be installed in final installation	N/A
4.9.4 Means to detect island situation	5.8.6 Islanding detection	The inverter has an active islanding detection capability but needs to be used with an external interface protection as dedicated device must be installed in final installation.	N/A
4.10.2 Automatic reconnection after tripping	5.9.3 Automatic reconnection after tripping	--	Pass
4.10.3 Starting to generate electrical power	5.9.4 Starting to generate electrical power	--	Pass
4.11.1 Ceasing active power	5.10 Active power reduction on set point	--	Pass
4.11.2 Reduction of active power on set point	5.10 Active power reduction on set point	--	Pass
4.12 Remote information exchange	5.11 Remote information exchange	Standardized communication protocol not provided by manufacturer	N/A

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Evaluated parameter and parameter range

<table>
<tr><td colspan="2">Specific technical requirement (e.g. grid codes)</td><td colspan="4">EN 50549-2:2019+A1:2023</td></tr>
<tr><td>Clause(s) / subclause(s) of EN 50549-2:2019+A1:2023</td><td>Parameter</td><td>Remarks/ additional information</td><td>Configurable value range</td><td colspan="2">Default value</td></tr>
<tr><td rowspan="6">4.4.2 Operating frequency range</td><td>47.0 - 47.5 Hz Duration</td><td>--</td><td>0 - 20 s</td><td colspan="2">20 s</td></tr>
<tr><td>47.5 - 48.5 Hz Duration</td><td>--</td><td>30 - 90 min</td><td colspan="2">30 min</td></tr>
<tr><td>48.5 - 49.0 Hz Duration</td><td>--</td><td>30 - 90 min</td><td colspan="2">30 min</td></tr>
<tr><td>49.0 - 51.0 Hz Duration</td><td>--</td><td>not configurable</td><td colspan="2">unlimited</td></tr>
<tr><td>51.0 - 51.5 Hz Duration</td><td>--</td><td>30 - 90 min</td><td colspan="2">30 min</td></tr>
<tr><td>51.5 - 52.0 Hz Duration</td><td>--</td><td>0 - 15 min</td><td colspan="2">15 min</td></tr>
<tr><td rowspan="2">4.4.3 Minimal requirement for active power delivery at underfrequency</td><td>Reduction threshold</td><td>--</td><td>not configurable</td><td colspan="2">No reduction.</td></tr>
<tr><td>Maximum reduction rate</td><td>--</td><td>not configurable</td><td colspan="2">Maximum allow 10 % of P_{max} per 1 Hz</td></tr>
<tr><td rowspan="2">4.4.4 Continuous operating voltage range</td><td>Upper limit</td><td>--</td><td>not configurable</td><td colspan="2">110% Un</td></tr>
<tr><td>Lower limit</td><td>--</td><td>not configurable</td><td colspan="2">90% Un</td></tr>
<tr><td>4.5.2 Rate of change of frequency (ROCOF) immunity</td><td>ROCOF withstand capability (defined with a sliding measurement window of 500 ms)</td><td>--</td><td>not configurable</td><td colspan="2">2.0 Hz/s</td></tr>
<tr><td rowspan="7">4.5.3.2 Under-voltage ride through (UVRT) Generating plant with non-synchronous generating technology</td><td>Maximum power resumption time</td><td>--</td><td>not configurable</td><td colspan="2">1 s</td></tr>
<tr><td rowspan="6">Voltage-Time-Diagram</td><td rowspan="2">--</td><td rowspan="6">See figure 6 most stringent curve of EN 50549-2:2019+A1:2023</td><td>Time[s]</td><td>U [p.u.]</td></tr>
<tr><td>--</td><td>0.05</td></tr>
<tr><td>--</td><td>0.25</td><td>0.05</td></tr>
<tr><td>--</td><td>3.0</td><td>0.85</td></tr>
<tr><td>--</td><td>180.0</td><td>0.85</td></tr>
<tr><td>--</td><td>180.0</td><td>0.90</td></tr>
<tr><td rowspan="4">4.5.4 Over-voltage ride through (OVRT)</td><td rowspan="4">Voltage-Time-Diagram</td><td rowspan="2">--</td><td rowspan="4">See figure 8 of EN 50549-2:2019+A1:2023</td><td>Time[s]</td><td>U [p.u.]</td></tr>
<tr><td>--</td><td>1.25</td></tr>
<tr><td>--</td><td>0.1</td><td>1.25</td></tr>
<tr><td>--</td><td>--</td><td>1.20</td></tr>
</table>

		--		5.0	1.20
		--		--	1.15
		--		60	1.15
		--		--	1.10
4.5.5 Phase jump immunity	Phase jump immunity	--	not configurable	50°	
4.6.1 Power response to overfrequency	Threshold frequency f1		50.2 Hz - 52 Hz	50.2 Hz	
	Droop	--	2 % - 12 %	5 %	
	Power reference	--	P_M | P_{max}	P_{max} for EESS	
	Intentional delay	--	0 - 2 s	0s	
	Deactivation threshold f_{stop}	--	50.0 Hz - f1	deactivated	
	Deactivation time t_{stop}	--	0 - 600 s	--	
	Acceptance of staged disconnection	--	yes | no	yes	
4.6.2 Power response to underfrequency	Threshold frequency f1	--	49.8 Hz - 46 Hz	49.8 Hz	
	Droop	--	2 - 12 %	5 %	
	Power reference	--	P_M | P_{max}	P_{max}	
	Intentional delay	--	0 - 2 s	0 s	
4.7.2.2 Voltage support by reactive power - Capabilities	Active factor / Reactive power (%P_n) range overexcited	--	0.9 - 1 / 0 kVar to 599.3 kVar	1 / 0 kVar	
	Active factor / Reactive power (%P_n) range underexcited	--	-0.9 - 1 / -599.3 kVar to 0 kVar	1 / 0 kVar	
4.7.2.3 Voltage support by reactive power - Control modes	Enabled control mode	--	Q setp Q(U) Cos φ setp Cos φ (P) Q (P)	Q setp(Q setpoint)	
4.7.2.3.2 Voltage support by reactive power - Setpoint control modes	Q setpoint and excitation	--	-599.3 kVar to 599.3 kVar	0 kVar	
	cos φ setpoint and excitation	--	0.9 underexcited - 0.9 overexcited	1	

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4.7.2.3.3 Voltage support by reactive power - Voltage related control modes	Characteristic curve - Q (U)	--	--	Indicate default characteristic
	Point a	--	90% - 100%Un	93%Un
	Point b	--	90% - 100%Un	94%Un
	Point c	--	100% - 110%Un	106%Un
	Point d	--	100% - 110%Un	108%Un
	Reactive power of Point a	--	-48%Pn - 48%Pn	48%Pn
	Reactive power of Point b	--	-48%Pn - 48%Pn	0%Pn
	Reactive power of Point c	--	-48%Pn - 48%Pn	0%Pn
	Reactive power of Point d	--	-48%Pn - 48%Pn	-48%Pn
	Time constant	--	3 s - 60 s	3 s
	Min cos φ	--	0.0 – 0.95	0.4
	Lock in power	--	0 % - 20 % P_n	deactivated
	Lock out power	--	0 % - 20 % P_n	deactivated
4.7.2.3.4 Voltage support by reactive power - Power related control mode	Characteristic curve - Cos φ(P) and Q(P)	--	--	Indicate default characteristic, Cos φ (P) and Q (P) can be enabled only one at a time
	Cos φ (P) Point a	--	0% - 100% P_n	15% P_n
	Cos φ (P) Point b	--	0% - 100% P_n	20% P_n
	Cos φ (P) Point c	--	0% - 100% P_n	80% P_n
	Cos φ (P) Point d	--	0% - 100% P_n	90% P_n
	Cos φ of Cos φ (P) Point a	--	0.9un - 0.9 ov	0.9 ov
	Cos φ of Cos φ (P) Point b	--	0.9un - 0.9 ov	1
	Cos φ of Cos φ (P) Point c	--	0.9un - 0.9 ov	1
	Cos φ of Cos φ (P) Point d	--	0.9un - 0.9 ov	0.9 un
	Q (P) Point a	--	0% - 100% P_n	15% P_n
	Q (P) Point b	--	0% - 100% P_n	20% P_n
	Q (P) Point c	--	0% - 100% P_n	80% P_n
	Q (P) Point d	--	0% - 100% P_n	90% P_n
	Reactive power of Q(P) Point a	--	-48%Pn - 48%Pn	48% P_n

	Reactive power of Q(P) Point b	--	-48%Pn - 48%Pn	0% P_n
	Reactive power of Q(P) Point c	--	-48%Pn - 48%Pn	0% P_n
	Reactive power of Q(P) Point d	--	-48%Pn - 48%Pn	-48% P_n
	Time constant	--	3 s - 60 s	3 s
	Lock in voltage	--	105%Un	deactivated
	Lock out voltage	--	100%Un	deactivated
4.7.3 Voltage related active power reduction	Characteristic curve - P (U)	--	--	Indicate default characteristic
	Voltage high Point 1	--	0%Un-135%Un	110%Un
	Voltage high Point 2	--	0%Un-135%Un	111%Un
	Voltage high Point 3	--	0%Un-135%Un	112%Un
	Power high Point 1	--	0% P_n - 100% P_n	100% P_n
	Power high Point 2	--	0% P_n - 100% P_n	50% P_n
	Power high Point 3	--	0% P_n - 100% P_n	0% P_n
	Time constant	--	3 s - 60 s	3 s
4.7.4.2.1 Voltage support during faults and voltage steps	Enabling	--	enable | disable	disabled
	Static voltage range overvoltage		100%Un-120%Un	110%Un
	Static voltage range undervoltage		80%Un-100%Un	90%Un
	Insensitivity range		0 % – 15 %	5 %
	Gradient k1		0 - 6	2
	Gradient k2		0 - 6	2
4.7.4.2.1.2 Optional Modes	Active power priority	--	enable | disable	This mode is not mandatory
	Reactive current limitation [% rated current]	--	0 %–100 %	This mode is not mandatory
	Zero current threshold	--	20%Un - 100%Un	This mode is not mandatory
4.7.4.2.2 Zero current mode for converter connected generating technology	Enabling	--	enable | disable	disable
	Static voltage range undervoltage		20%Un - 100%Un	50%Un

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4.9.3 Requirements on voltage and frequency protection	Threshold for protection as dedicated device [in A or kW, kVA]	--	16 A - 250 kVA	External interface protection for generating plants to be connected to the MV network according to EN 50549-2:2019+A1:2023 must be installed in final installation
	Undervoltage threshold stage 1	--	20%Un - 100%Un	--
	Undervoltage operate time stage 1	--	0.1 s -100 s	--
	Undervoltage threshold stage 2	--	20%Un - 100%Un	--
	Undervoltage operate time stage 2	--	0.1 s - 5 s	--
	Overvoltage threshold stage 1	--	100%Un-120%Un	--
	Overvoltage operate time stage 1	--	0.1 s -100 s	--
	Overvoltage threshold stage 2	--	100%Un-130%Un	--
	Overvoltage operate time stage 2	--	0.1 s - 5 s	--
	Overvoltage threshold 10 min mean protection	--	100%Un-115%Un	--
	Underfrequency threshold stage 1	--	47.0 Hz - 50.0 Hz	--
	Underfrequency operate time stage 1	--	0.1 s -100 s	--
	Underfrequency threshold stage 2	--	47.0 Hz - 50.0 Hz	--
	Underfrequency operate time stage 2	--	0.1 s - 5 s	--
	Overfrequency threshold stage 1	--	50.0 Hz - 52.0 Hz	--
	Overfrequency operate time stage 1	--	0.1 s -100 s	--
	Overfrequency threshold stage 2	--	50.0 Hz - 52.0 Hz	--
	Overfrequency operate time stage 2	--	0.1 s - 5 s	--

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	Positive sequence under-voltage protection threshold	--	20%Un - 100%Un	--
	Positive sequence under-voltage protection operate time	--	0.2 s -100 s	--
	Negative sequence over-voltage protection threshold	--	1%Un - 100%Un	--
	Negative sequence over-voltage protection threshold time	--	0.2 s -100 s	--
	Zero sequence over-voltage protection threshold	--	1%Un - 100%Un	--
	Zero sequence over-voltage protection threshold time	--	0.1 s -100 s	--
4.10.2 Automatic reconnection after tripping	Lower frequency	--	47.0 Hz - 50.0 Hz	49.5 Hz
	Upper frequency	--	50.0 Hz - 52.0 Hz	50.2 Hz
	Lower voltage	--	50%Un-100%Un	90%Un
	Upper voltage	--	100%Un-120%Un	110%Un
	Observation time	--	10 s - 600 s	60 s
	Active power increase gradient	--	0.1%/s - 50%/s	0.16%/s
4.10.3 Starting to generate electrical power	Lower frequency	--	47.0 Hz - 50.0 Hz	49.5 Hz
	Upper frequency	--	50.0 Hz - 52.0 Hz	50.1 Hz
	Lower voltage	--	50%Un-100%Un	90%Un
	Upper voltage	--	100%Un-120%Un	110%Un
	Observation time	--	10 s - 600 s	60 s
	Active power increase gradient	--	0.1%/s - 50%/s	disabled
4.11.1 Ceasing active power	Activation option	--	Can be achieved by BCS: WEB1/2/3/4 port via Modbus communication protocol, decision should be made by the DSO and responsible party	
4.11.2 Reduction of active power on set point	Activation option	--	Can be achieved by BCS: WEB1/2/3/4 port via Modbus communication protocol, decision should be made by the DSO and responsible party	

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4.12 Remote information exchange	Available communication standards	--	Standardized communication protocol not provided by manufacturer